

Movie Rental Data Warehouse Project

Final ETL and Dimensional Modeling Report

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Course : Data_Base 2

1. Introduction

This project presents the design and implementation of a Movie Rental Data Warehouse based on the Sakila database. The goal of the project is to transform operational rental transaction data into an analytical environment that supports reporting and business intelligence operations.

The warehouse was designed using dimensional modeling techniques including fact tables and dimension tables. ETL processes were implemented to extract data from the operational database, transform it into analytical structures, and load it into the warehouse.

The developed warehouse enables efficient analytical queries related to customers, rentals, revenues, stores, and film performance.

2. Business Questions

The data warehouse was designed to answer several important analytical business questions, including:

- Which customers generate the highest total payments?
- Which films produce the highest revenue?
- What are the busiest rental periods?
- Which store achieves better rental performance?
- How does film revenue vary over time?

These questions support management decisions and business performance evaluation.

Business Question	Main Fact Table	Supporting Dimensions
Which customers generate the highest total payments?	fact payment	dim customer, dim date
Which films produce the highest revenue?	fact film performance	dim film, dim date
What are the busiest rental periods?	fact rental	dim date
Which store achieves better rental performance?	fact rental, fact payment	dim store, dim date

How does film revenue vary over time?	fact film performance	dim film, dim date
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3. Dimensional Model Design

3.1 Selected Business Processes

The following business processes were selected for the warehouse design:

- Payment Transactions
- Rental Transactions
- Film Performance Analysis

Each process was modeled using a fact table connected to multiple dimensions.

3.2 Fact Tables

Three fact tables were implemented:

Fact Table	Description
Fact_payment	Stores payment transactions including customer, staff, store, date, and payment amount.
Fact_rental	Stores rental transaction details and rental counts.
Fact_film _performance	Stores aggregated film rental statistics and total generated revenue.

Fact Table	Business Process	Grain	Measures	Related Dimensions
Fact _payment	Payment Transactions	One row per payment transaction	amount	dim _customer, dim_staff, dim_store, dim_date
fact _rental	Rental Transactions	One row per rental transaction	rental count	dim_customer, dim_film, dim_staff, dim_store, dim_date
fact_film_performance	Film Performance Analysis	One row per film per date	total rentals, total revenue	dim_film, dim_date

3.3 Dimension Tables

The warehouse contains several dimensions:

- dim customer

- dim_film
- dim_store
- dim_staff
- dim_date

These dimensions provide descriptive business context for analytical operations.

DW Table	Source OLTP Tables	Purpose
dim_customer	customer, address, city, country	Stores customer descriptive information and location.
dim_film	film, film category, category	Stores film details and category information.
dim_store	store, address, city, country	Stores store and location information.
dim_staff	staff, store	Stores staff information and assigned store.
dim_date	rental, payment dates	Supports time-based analysis by day, month, quarter, and year.
fact_payment	payment, customer, staff, rental, inventory	Stores payment transactions and revenue measures.
fact_rental	rental, inventory, customer, staff, film	Stores rental transactions and rental count measures.
fact_film_performance	rental, inventory, film, payment	Stores aggregated film rental and revenue performance.

4. Dimensional Model Diagram

The model follows a star schema approach. The fact tables store numeric business measures, while the dimension tables provide descriptive context such as customer, film, store, staff, and date information.

Shared dimensions such as dim date, dim customer, dim film, dim store, and dim staff allow the warehouse to answer multiple analytical questions.

Figure below shows the final dimensional model and relationships between fact and dimension tables.

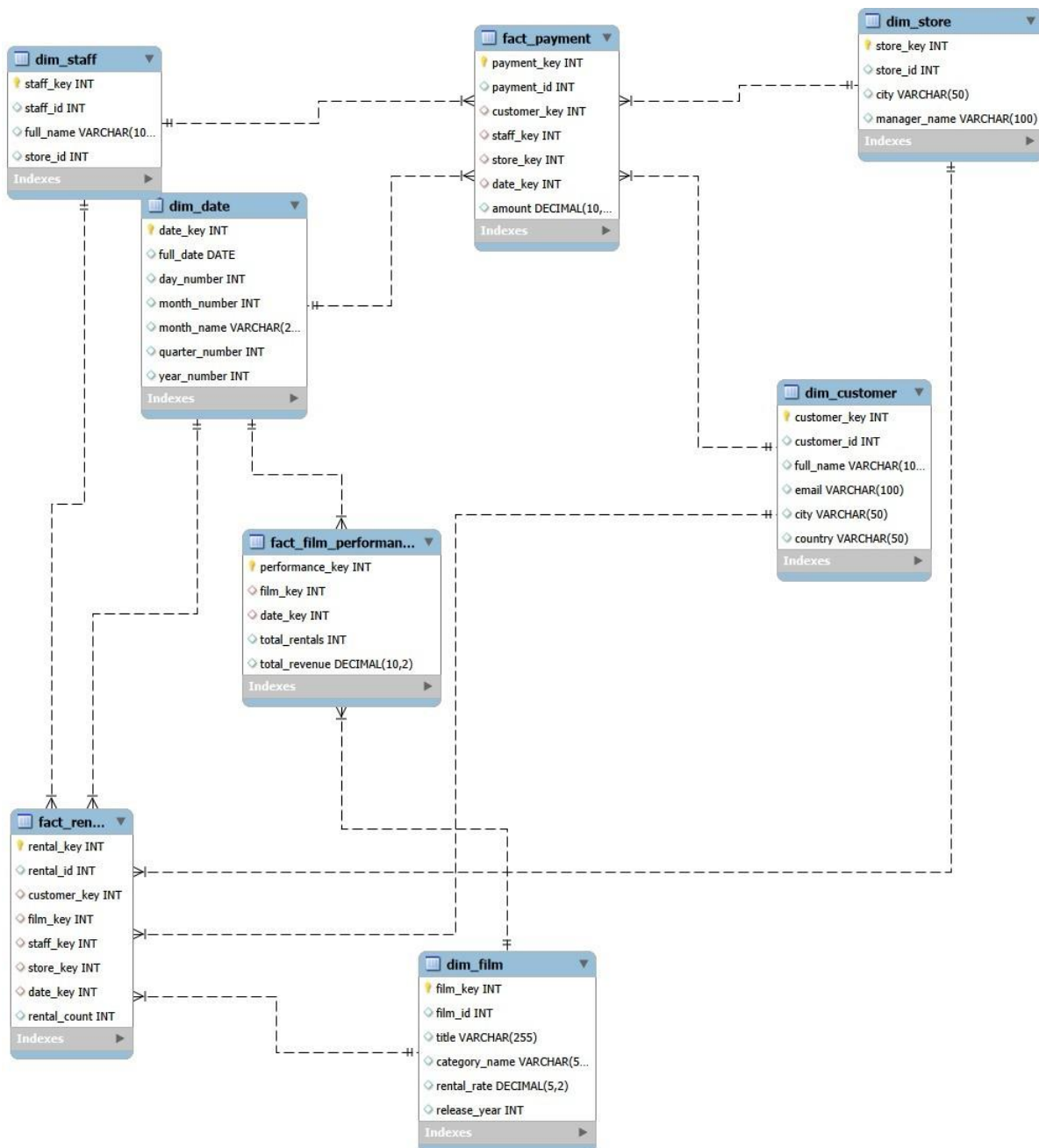


Figure 1: Final Data Warehouse Schema

5. ETL Design

5.1 Extract

The ETL process extracts operational data from the Sakila OLTP database. The main extracted tables are rental, payment, customer, film, inventory, store, staff, address, city, country, category, and film category.

These tables are required because they contain the rental and payment transactions, as well as the descriptive information needed to build customer, film, store, staff, and date dimensions.

5.2 Transform

During transformation, related OLTP tables are joined to create analytical dimensions. Customer first and last names are combined into full names. Film data is enriched with category information. Store and customer locations are built by joining address, city, and country tables.

Operational IDs are converted into warehouse surrogate keys. Date values are transformed into date keys, month, quarter, and year attributes.

Measures such as payment amount, rental count, total rentals, and total revenue are calculated. Missing or invalid values are checked before loading.

5.3 Load

Dimension tables are loaded before fact tables because fact tables depend on dimension surrogate keys.

After the dimensions are populated, fact tables are loaded by looking up the correct customer, film, store, staff, and date keys.

Data quality checks are applied before and after loading to prevent duplicate records, missing foreign keys, and invalid measure values.

6. Data Quality Considerations

To improve warehouse quality and consistency, several considerations were applied:

- Duplicate prevention
- Referential integrity validation
- Null value handling
- Data type consistency
- Foreign key relationship validation
- Validation of positive payment values
- Detection of invalid rental dates

- Standardization of customer and location names

These practices ensure accurate and reliable analytical results.

7. Sample Analytical Queries

The following examples show how the data warehouse can answer the selected business questions.

Query 1: Top Customers by Total Payments

```
SELECT dc.full_name,  
       dc.city, dc.country,  
       SUM(fp.amount) AS total_payment  
FROM fact_payment fp  
JOIN dim_customer dc  
     ON fp.customer_key = dc.customer_key  
GROUP BY dc.full_name,  
         dc.city, dc.country  
ORDER BY total_payment DESC  
LIMIT 10;
```

Query 2: Top Films by Rentals

```
SELECT df.title,  
       df.category_name,  
       SUM(fr.rental_count) AS total_rentals  
FROM fact_rental fr  
JOIN dim_film df  
     ON fr.film_key = df.film_key  
GROUP BY df.title,  
         df.category_name  
ORDER BY total_rentals DESC  
LIMIT 10;
```

Query 3: Monthly Revenue

```

SELECT dd.year_number,
       dd.month_number,
       dd.month_name,
       SUM(fp.amount) AS monthly_revenue
FROM fact_payment fp
JOIN dim_date dd
     ON fp.date_key = dd.date_key
GROUP BY dd.year_number,
         dd.month_number,
         dd.month_name
ORDER BY dd.year_number,
         dd.month_number;

```

8. Analytical Visualization

The visualization below shows the top customers based on total spending. This supports the business question about which customers generate the highest total payments.

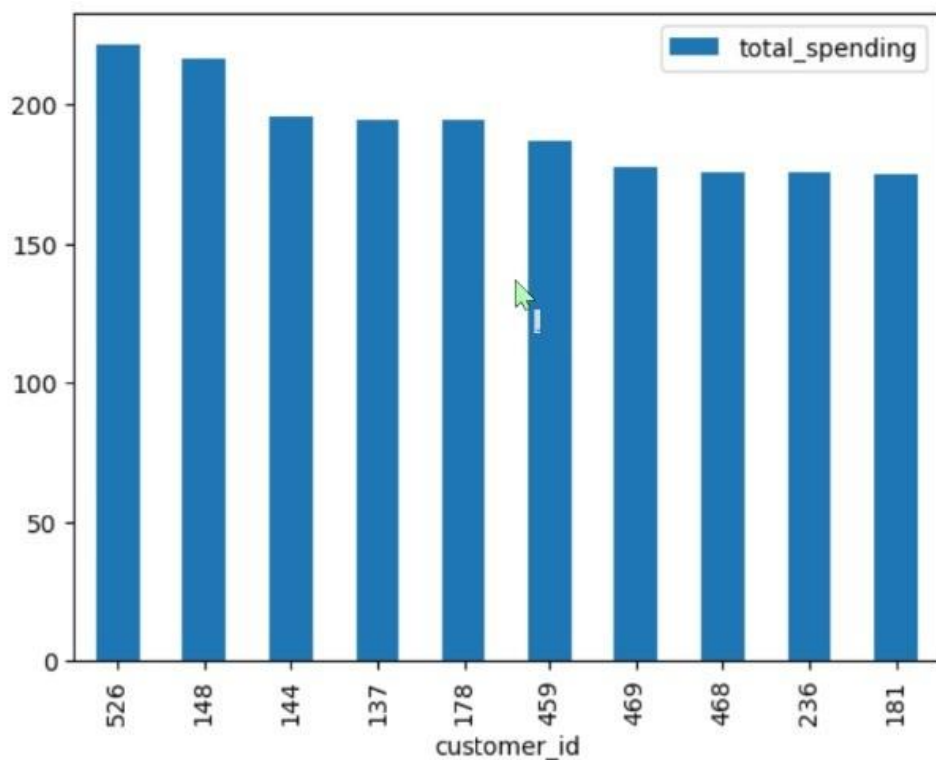


Figure 2: Top Customers by Total Spending

9. Conclusion

This project successfully designed and implemented a Movie Rental Data Warehouse prototype using dimensional modeling and ETL methods.

The warehouse structure supports analytical reporting and provides useful business insights about customers, films, stores, revenues, and rental activities.

The final solution shows how operational transaction data from the Sakila database can be transformed into an analytical structure suitable for business intelligence and decision-making.